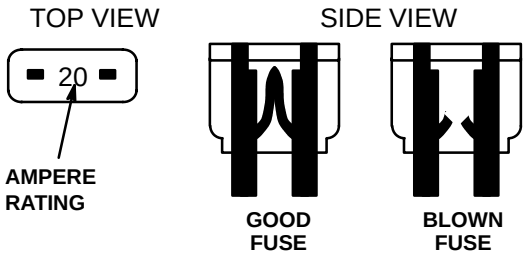
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

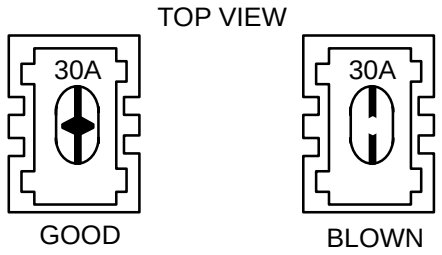


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

MAXI-FUSE



Cartridge maxi-fuses have transparent plastic tops. To check a cartridge maxi-fuse, look at the fuse element through the top of the housing.

To replace a cartridge maxi-fuse, pull it from the fuse box or panel. Always replace a blown cartridge maxi-fuse with a new one having the same ampere rating.

The ampere rating of a cartridge maxi-fuse can also be determined by following the color code shown here:

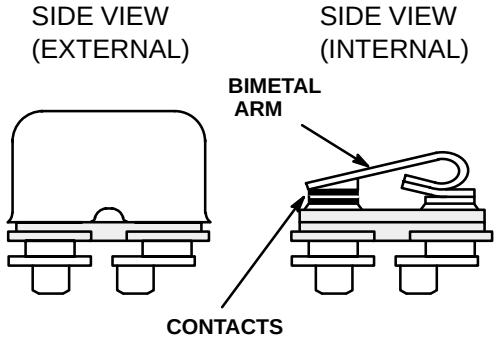
CARTRIDGE TYPE FUSIBLE LINKS	
High Current Fuse Value Amps	**Color Code**
20A Cartridge	Blue
30A Cartridge	Pink
40A Cartridge	Green
50A Cartridge	Red

CIRCUIT BREAKER

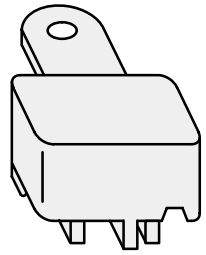
Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart). They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE



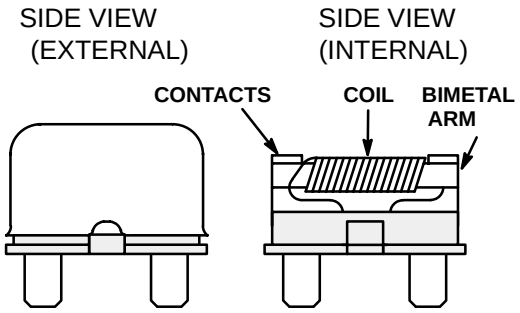
IN-LINE MOUNTED CYCLING TYPE



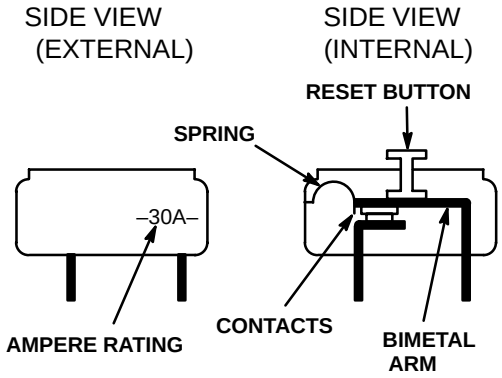
In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

FUSE PANEL MOUNTED

NON-CYCLING TYPE



FUSE PANEL MOUNTED
MANUAL RESET TYPE

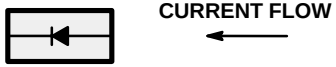


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



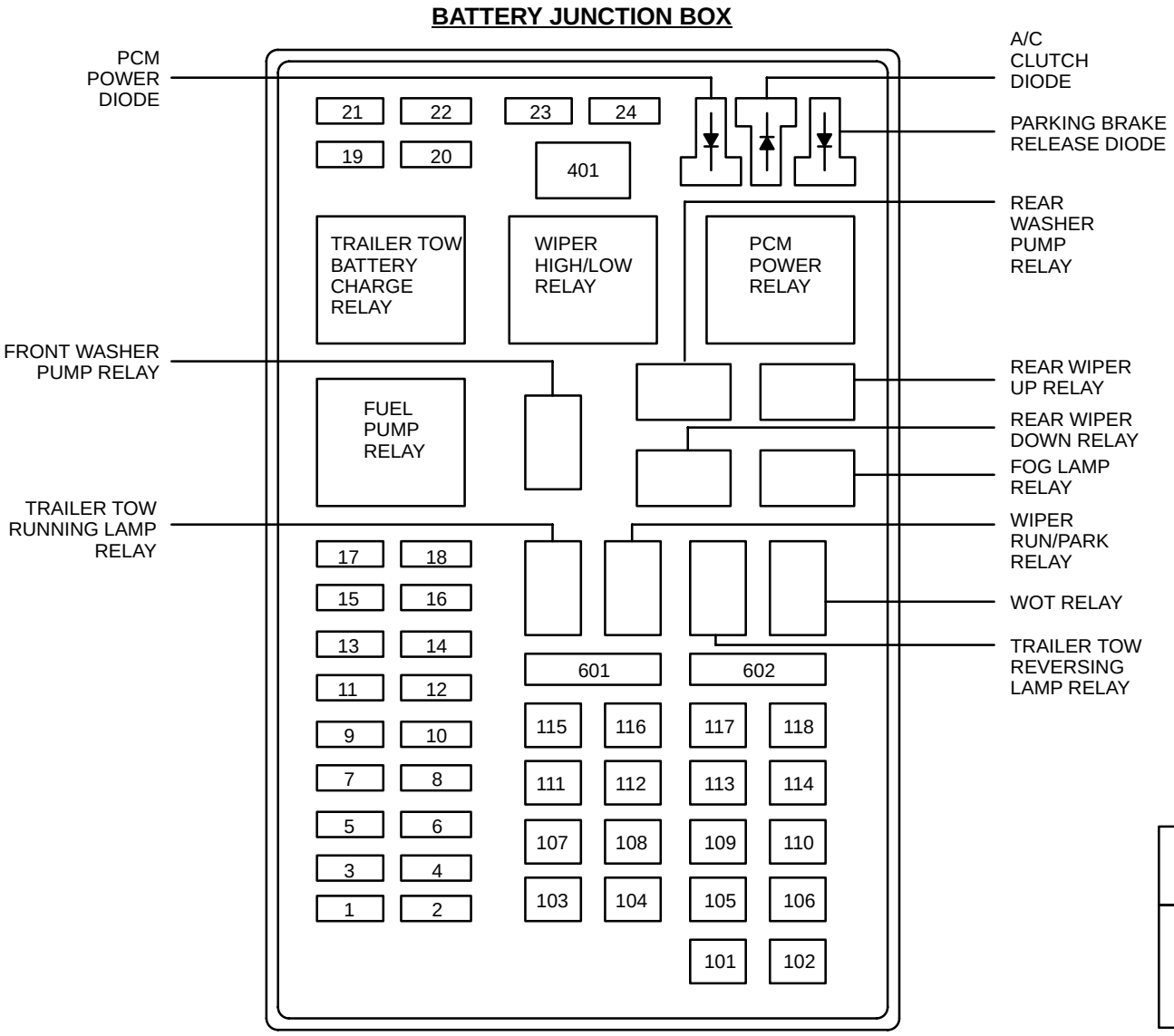
Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

MEGA FUSE

TOP VIEW

Mega fuse protects the charging circuit. In the event that the mega fuse is blown due to the charging circuit failure, the generator field circuit is disabled.

11-3 FUSE PANEL/CIRCUIT PROTECTION



HIGH CURRENT FUSE VALUE AMPS	COLOR CODE
20A PLUG-IN	YELLOW
30A PLUG-IN	GREEN
40A PLUG-IN	ORANGE
50A PLUG-IN	RED
60A PLUG-IN	BLUE

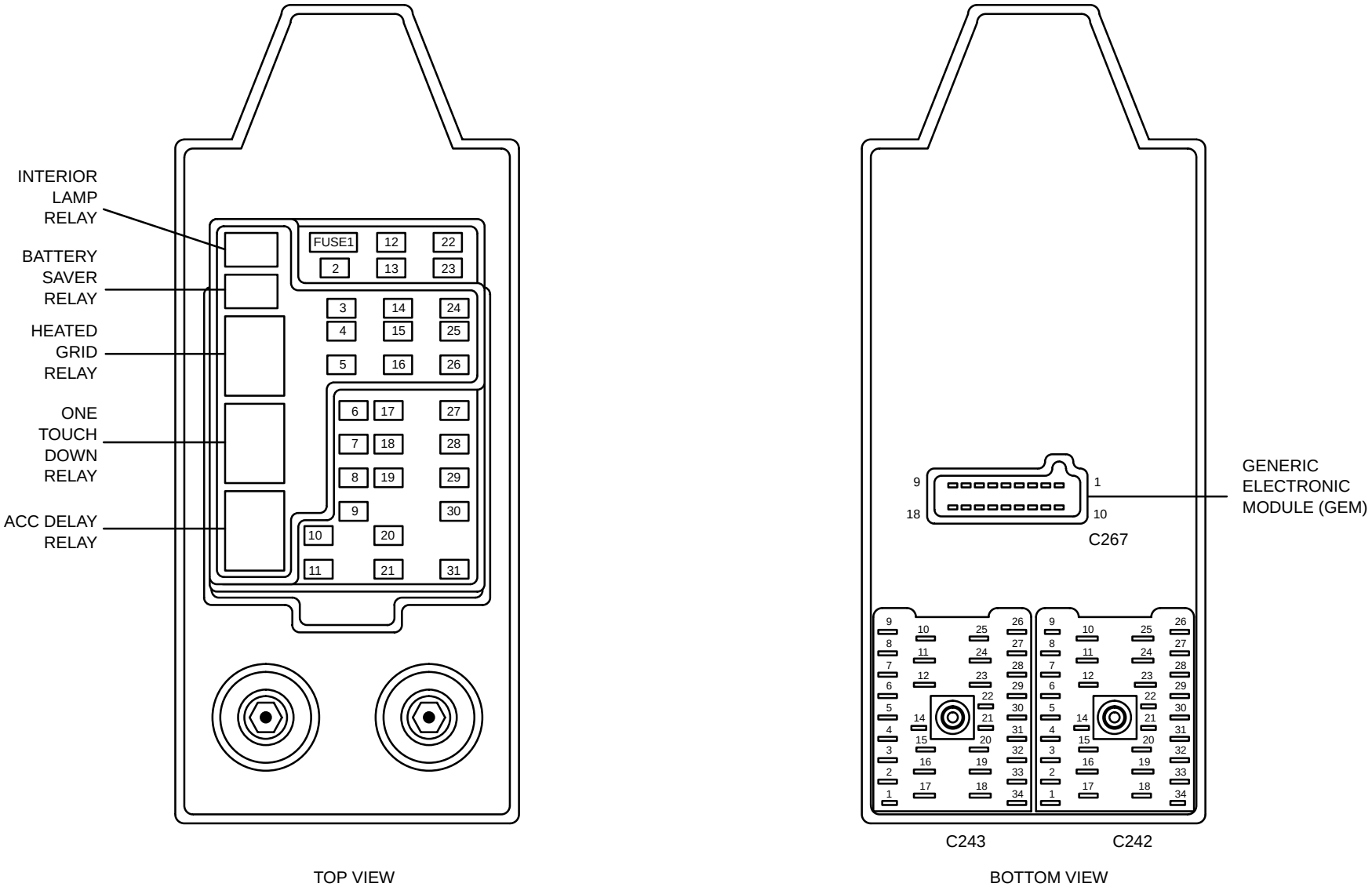
BATTERY JUNCTION BOX

Mini Fuses	Amps	Circuits Protected
1	20	Auxiliary Power Socket
2	30	PCM Power Relay
3	30	Main Light Switch, Headlamp Relay, Multifunction Switch
4	20	Console Auxiliary Power Socket
5	20	Trailer Tow Running Lamp Relay, Trailer Tow Reversing Lamp Relay
6	15	Main Light Switch, Central Security Module
7	20	Central Security Module
8	30	Central Security Module
9	15	Daytime Running Lamps (DRL) Resistor, Fog Lamp Relay
10	20	Fuel Pump Relay
11	20	Charging System
12	10	Rear Wipers Up and Down
13	15	WOT Relay
14	–	NOT USED
15	10	Right Running Board Lamp, Left Running Board Lamp
16	–	NOT USED
17	10	Radio, Vent Windows
18	15	Fuel Injectors, PCM
19	–	NOT USED
20	–	NOT USED
21	–	NOT USED
22	–	NOT USED
23	15	Heated Oxygen Sensors, Transmission
24	–	NOT USED

Maxi Fuses	Amps	Circuits Protected
101	30	Trailer Tow Battery Charge Relay
102	50	(4WABS) Module
103	50	Central Junction Box
104	30	Transfer Case Shift Relays, Torque On Demand (TOD) Relay
105	40	Blower Relay
106	–	NOT USED
107	30	Passenger Power Seat Control Switch, Passenger Power Lumbar Switch
108	30	Trailer Tow (Battery Feed)
109	50	Air Suspension Compressor Relay
110	30	Temperature Controlled Seat Modules, Heated Seat Modules
111	40	Ignition Switch (B4 and B5)
112	30	Driver Power Seat Control Switch, Driver Power Lumbar Switch, Seat Module, Adjustable Pedal Switch
113	40	Ignition Switch (B1 and B3)
114	30	Auxiliary A/C Relay
115	–	NOT USED
116	40	Heated Grid Relay, Heated Mirrors
117	–	NOT USED
118	–	NOT USED
601 c.b.	30	Acc Delay Relay to Windows and Sliding Roof Panel
602 c.b.	–	NOT USED

11-5 FUSE PANEL/CIRCUIT PROTECTION

CENTRAL JUNCTION BOX



CENTRAL JUNCTION BOX

Fuse Position	Amps	Circuits Protected
1	25	Radio, Subwoofer Amplifier
2	5	Instrument Cluster, Overhead Trip Computer (OTC) Module, Powertrain Control Module (PCM), Electronic Automatic Temperature Control (EATC), Clock, Navigation
3	20	Cigar Lighter, Data Link Connector (DLC)
4	7.5	Power Mirror Switch, Memory System, Central Security Module
5	15	Digital Transmission Range (DTR) Sensor, Speed Control Servo, Function Selector Switch, Blend Door Actuator, Electronic Variable Orifice (EVO) Module, Daytime Running Lamps (DRL) Relay Block
6	5	Shift Lock Actuator, Generic Electronic Module (GEM), Air Suspension/EVO Steering Module, Compass Sensor, Steering Wheel Rotation Sensor, Heated Grid Relay, Overhead Trip Computer (OTC) Module, Instrument Cluster, Reverse Sensing System
7	5	Auxiliary A/C Relay, Console Blower Relay
8	5	Radio, Remote Anti-Theft Personality (RAP) Module, Clock, Cellular Phone Support Electronics, Navigation, Rear Seat Entertainment
9	–	NOT USED
10	–	NOT USED
11	30	Washer Pump Relay, Wiper Run/Park Relay, Wiper HI/LO Relay, Windshield Wiper Motor, Rear Washer Pump Relay
12	15	Air Suspension Service Switch
13	20	Brake Pedal Position (BPP) Switch, Brake Pressure Switch, Flasher Relay
14	15	Battery Saver Relay, Interior Lamp Relay
15	5	Generic Electronic Module (GEM), Brake Pedal Position (BPP) Switch
16	20	Instrument Cluster, High Beam Headlamps (Power supplied through Multifunction Switch)
17	10	Heated Backlite Switch, Left Exterior Rear View Mirror, Right Exterior Rear View Mirror
18	5	Main Light Switch, Instrument Illumination (Power supplied through Main Light Switch), Clock, Navigation
19	–	NOT USED
20	5	Air Suspension/EVO Steering Module, Generic Electronic Module (GEM), Driver Seat Module, Radio
21	15	Digital Transmission Range (DTR) Sensor, Central Junction Box (Fuse 20)
22	10	Air Bag Diagnostic Monitor, Function Selector Switch, Blower Relay, Electronic Automatic Temperature Control (EATC), Central Junction Box (Fuse 7)
23	10	Trailer Tow Battery Charge Relay, Auxiliary A/C Blend Actuator, Electronic Day/Night Mirror, Rear Integrated Control Panel, Heated Seats, Auxiliary A/C Mode Actuator, Auxiliary A/C Control Module, Flasher Relay, Torque On Demand (TOD) Relay, 4 Wheel Anti-Lock Brake System (4WABS) Module
24	10	Blower Motor Relay, Electronic Automatic Temperature Control (EATC) Module, Function Selector Switch Assembly
25	–	NOT USED
26	10	Right Headlamp (Power Supplied Through Multifunction Switch)
27	5	Main Light Switch, Fog Lamp Relay
28	10	Left Headlamp (Power Supplied Through Multifunction Switch)
29	5	Autolamp Module, Transmission Control Switch (TCS), Belt Minder Module
30	30	Radio Noise Capacitor, Ignition Coil, PCM Power Diode, Coil On Plugs, Anti-Theft Transceiver Module, Instrument Cluster
31	10	Rear Integrated Control Panel (RICP) (To Audio), CD Player, Cellular Phone